Operations Research (MTH601)

Question: 1 (Marks: 1) Attempted Questions: 0 Total Questions: 50

The cost matrix in assignment problem is always diagonal matrix





Choices:

square matrix

identity matrix

zero matrix

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Verify Answers Correct as necessary. Do not blindly copy and Paste Answers given in this document.

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Question: 2 (Marks: 1) Attempted Questions: 1 Total Questions: 50

During a replacement if the value of money decreases at the rate of 3% then the present worth factor of unit amount to be spent after one year is given by

Choices:

- 0.25
- 0.333
- 0.9708
- 4

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Question: 3 (Marks: 1) Attempted Questions: 2 Total Questions: 50

Transportation problem is a special class of linear programming problem in which transport a commodity from the source to a destination in such a way that the total transportation cost is

.....

Choices:

Zero

.....

Maximum

.....

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Minimum

.....

Average

.....

Question: 4 (Marks: 1) Attempted Questions: 3 Total Questions: 50

In assignment problem, each job requires exactly one resource.

Choices:

True

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False

May not be true

Question: 5 (Marks: 1) Attempted Questions: 4 Total Questions: 50

In assignment problem each resource can be assigned

.....

Choices:

exactly one job

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.....

At least one job

.....

At most one job

.....

Question: 6 (Marks: 1) Attempted Questions: 6 Total Questions: 50

In assignment problem, each job requires exactly one resource.

Choices:

True

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False

May not be true

Question: 7 (Marks: 1) Attempted Questions: 6 Total Questions: 50

We go in probabilistic replacement model when period between installation and failure is

.....

Choices:

varying exponentially

.....

varying linearly

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.....

Constant

.....

is not constant (varying arbitrarily)

.....

Question: 8 (Marks: 1) Attempted Questions: 7 Total Questions: 50

The present worth of a rupee to be spent after a year is denoted by v and given by

Choices:

- $v = 1/(1+r)$
- $v = (1+r)$
- $v = (1/r)$
- $v = (1+r)/10$

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Question: 9 (Marks: 1) Attempted Questions: 8 Total Questions: 50

In replacement Models, Capital Cost = (----------) – (Resale Cost).

Choices:

- Item / Machine Cots
- Running Cots
- Maintenance Cots
- Average Cost

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Question: 10 (Marks: 1) Attempted Questions: 9 Total Questions: 50

A non-positive solution which doesn't satisfy all the constraints of linear programming problem is known as -----

Choices:

- Primal
- Feasible
- Non-feasible
- Optimal

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Question: 12 (Marks: 1) Attempted Questions: 11 Total Questions: 50

In the simplex table for a linear programming problem, we select the leaving basic variable corresponds to ----------

Choices:

- Maximum non-negative ratio
- Maximum negative ratio
- Minimum non-negative ratio
- Minimum negative ratio

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Question: 11 (Marks: 1) Attempted Questions: 12 Total Questions: 50

While using Simplex method for a LPP, if there are 4 standard equations in 8 variables, then the number of non-basic variables will be -------

Choices:

32

12

4

2

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Question: 13 (Marks: 1) Attempted Questions: 13 Total Questions: 50

In Two – Phase method, if all the _____ variables become zero, we stop the phase I and proceed to phase-II.

Choices:

- artificial
- basic
- non basic
- negative slack
- positive slack

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Question: 14 (Marks: 1) Attempted Questions: 13 Total Questions: 50

In case of degeneracy, a linear programming problem with two different bases has
---------- feasible solution.

Choices:

- unbounded
- two distinct
- multiple
- the same

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Question: 15 (Marks: 1) Attempted Questions: 14 Total Questions: 50

In a temporary degenerate solution, any basic variable in the initial iteration becomes ---------, while in the later iterations becomes--------.

Choices:

positive, zero

zero, positive

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zero,infinte

infinte,zero

Question: 16 (Marks: 1) Attempted Questions: 15 Total Questions: 50

Due to which of the following reason, unboundedness is recognized in maximizing a linear programming problem?

Choices:

Some of the cost coefficients of Objective function are zero or negative

Some of the cost coefficients of Objective function are zero or positive

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All the constraint coefficients of any non-basic variables are zero or negative

All the constraint coefficients of any non-basic variables are zero or positive

Question: 17 (Marks: 1) Attempted Questions: 16 Total Questions: 50

In case of alternative or multiple optima, objective function line is ---------- to any binding constraint.

Choices:

perpendicular

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inclined

parallel

Question: 18 (Marks: 1) Attempted Questions: 17 Total Questions: 50

_____ primal variable always corresponds to an equality dual constraint.

Choices:

An artificial

A basic

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A non-basic

An unrestricted

Question: 19 (Marks: 1) Attempted Questions: 18 Total Questions: 50

Transportation problem is a special class of linear programming problem in which transport a commodity from the source to a destination in such a way that the total transportation cost is -------.

Choices:

Zero

Maximum

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Minimum

Average

Question: 20 (Marks: 1) Attempted Questions: 19 Total Questions: 50

Many decision making problems involve a process that takes place in multiple stages in such a way that at each stage, the process is dependent on the strategy chosen. Such types of problems are called

Choices:

- Linear Programming problems
- Integer programming problems
- Dynamic programming problems
- Assignment problems

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Question: 21 (Marks: 1) Attempted Questions: 20 Total Questions: 50

In sequencing problems, the Johnson's algorithm in finding the optimal ordering of n jobs through 3 machines can be applied, if the problem is converted into following number of machines problems

Choices:

- $3n$
- $3n!$
- $2 \times 3 = 6$
- 2

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Question: 22 (Marks: 1) Attempted Questions: 21 Total Questions: 50

The representation of reality in some physical form or in some form of mathematical equations may be discussed under the topic entitled as

Choices:

- Transportation problems
- Game theory
- Simulation
- Replacement problems

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Question: 23 (Marks: 1) Attempted Questions: 22 Total Questions: 50

The initial feasible solution obtained by the Least Cost method usually gives the -------- transportation cost than the obtained in North-West corner.

Choices:

- same
- less
- more
- very high

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Question: 24 (Marks: 1) Attempted Questions: 23 Total Questions: 50

In dynamic Programming the possible decisions at any stage, from which we are to choose one, are called states.

Choices:

True

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False

Question: 25 (Marks: 1) Attempted Questions: 24 Total Questions: 50

A problem in which the decisions have to be made at successive stages ,is called a multistage decision problem.

Choices:

True

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False

Question: 26 (Marks: 1) Attempted Questions: 25 Total Questions: 50

Which of the following is the reason to employ the queuing theory?

Choices:

- To generate more arrivals to the system
- To reduce service times
- To reduce customer wait time in a line
- None of these

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Question: 27 (Marks: 1) Attempted Questions: 26 Total Questions: 50

Which of the following distribution is most often used to estimate the arrival pattern?

Choices:

- Bernoulli
- Binomial
- Normal
- Poisson

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Question: 28 (Marks: 1) Attempted Questions: 27 Total Questions: 50

Integer programming problem is a special case of L.P.P where all or some variables are constrained to assume non-negative integer values.

Choices:

True

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False

Question: 29 (Marks: 1) Attempted Questions: 28 Total Questions: 50

A stochastic process is a .....

Choices:

Probabilistic process

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Deterministic process

Question: 30 (Marks: 1) Attempted Questions: 29 Total Questions: 50

Transportation problem is said to be unbalanced if .....

.....

Choices:

- Total supply is not equal to total demand.

.....

- Total supply is equal to total demand.

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.....

- Total supply is greater than total demand.

.....

- Total supply is lesser than total demand.

.....

Question: 31 (Marks: 1) Attempted Questions: 30 Total Questions: 50

The most common assumption in the queue theory is .....

Choices:

First In First Out

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Last In First Out

Question: 32 (Marks: 1) Attempted Questions: 31 Total Questions: 50

In dynamic programming problems, the main problem is divided into sub-problems. Each sub-problem is known as_____.

Choices:

- Part
- Stage
- State
- Mini problem

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Question: 33 (Marks: 1) Attempted Questions: 32 Total Questions: 50

For a Poisson random variable, λ represents the _____ number of arrivals per period time.

Choices:

- Maximum
- Mean
- Minimum
- None of these

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Question: 34 (Marks: 1) Attempted Questions: 33 Total Questions: 50



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Which of the following inequality corresponds to above shaded unbounded feasible region?

Choices:

- $x + 2y \leq 3$
- $x + 2y \geq 3$
- $x - 2y \geq 3$
- $x - 2y \leq 3$

For the Linear Programming problem,

$Max\ Z = 5x_1 + 6x_2$

Subject to

x_1 *is unrestricted*, $x_2 \geq 0$

Which of the following is the Standard form of its Primal's objective function?

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Choices:

- $Max\ Z = 5x_1^+ + 5x_1^- + 6x_2$
- $Max\ Z = 5x_1 + 6x_2$
- $Max\ Z = 5x_1^+ + 5x_1^- + 6x_2^+ + 6x_2^-$
- $Max\ Z = 5x_1^+ - 5x_1^- + 6x_2$

Question: 36 (Marks: 1) Attempted Questions: 35 Total Questions: 50

While using Least Cost Transportation Method for the table below, in which of the following cell the first allocation will be made?

Origins	P	Q	R	Supply
A	1	2	3	3
B	4	3	2	4
C	0	2	2	5
Demand	4	2	6	

Choices:

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- (R, B) or (R, C)
- (Q, C) or (Q, A)
- (P, A)
- (P, C)

Question: 37 (Marks: 1) Attempted Questions: 36 Total Questions: 50

The constrained $3x_1 + 2x_2 \leq 18$ in standard form can be written as:
(Where R is artificial variable and S1 is Slack variable)

Choices:

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- $3x_1 + 2x_2 + S_1 = 18$
- $3x_1 + 2x_2 + S_1 - R = 18$
- $3x_1 + 2x_2 + S_1 + R = 18$
- $3x_1 + 2x_2 - S_1 + R = 18$

Question: 38 (Marks: 1) Attempted Questions: 37 Total Questions: 50

Optimality test to any feasible solution of a $(m \times n)$ transportation problem can be performed if it fulfills the following conditions.

I. It consists of exactly (m + n – 1) individual cells being allocated.
II. These allocations are in independent positions.

Choices:

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True

False

Consider the initial basic feasible solution using Least Cost Method. For finding the optimal solution: Set

$u_1 = 0, v_2 = 1, v_3 = 2$ *and* $u_2 = 2$, what is the value of v_1?

	D1	D2	D3	Supply	
S1	5	1 (20)	2 (5)	25	u_1
S2	7	2	4 (10)	10	u_2
S3	3 (10)	3	5 (5)	15	u_3
Demand	10	20	20		
	v_1	v_2	v_3		

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Choices:

- 3
- 2
- 0
- 4

Question: 40 (Marks: 1) Attempted Questions: 39 Total Questions: 50

Consider the initial basic feasible solution of a Transportation problem by North – West Corner Method.
Finding optimal basic feasible solution: Suppose $u_1 = 0, v_1 = 5, v_2 = 1, u_2 = 1$ *and* $v_3 = 3$, what is the value of u_3?

	D1	D2	D3	Supply	
S1	5 (10)	1 (15)	2	25	u_1
S2	7	2 (5)	4 (5)	10	u_2
S3	3	3	5 (15)	15	u_3
Demand	10	20	20		
	v_1	v_2	v_3		

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Choices:

-
-
-
-

Consider the initial basic feasible solution of a Transportation problem by North – West Corner Method.
Finding optimal basic feasible solution: Suppose $u_1 = 0, v_1 = 5, v_2 = 1, u_2 = 1$ *and* $v_3 = 3$, what is the value of u_3?

	D1	D2	D3	Supply	
S1	5 (10)	1 (15)	2	25	u_1
S2	7	2 (5)	4 (5)	10	u_2
S3	3	3	5 (15)	15	u_3
Demand	10	20	20		
	v_1	v_2	v_3		

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Choices:

- 3
- 5
- 2
- 1

Markets / salesmen	I	II	III	IV
A	44	80	52	60
B	60	56	40	72
C	36	60	48	48
D	52	76	36	40

Operate first and second operation (row & column) by optimize the above assignment problem.

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Answer:

Question: 42 (Marks: 3) Attempted Questions: 40 Total Questions: 50

The cost of a new machine is Rs. 5000. The maintenance cost during the nth year is given by M_n = Rs. 500 (n – 1), where n = 1, 2, 3, ---. If the discount rate per year is 0.05, determine discount cost ($R_t v^{t-1}$) for each year.

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Answer:

Question: 43 (Marks: 3) Attempted Questions: 41 Total Questions: 50

Describe the advantages of the Duality Theory to solve a Linear Programming Problem.

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Answer:

Question: 45 (Marks: 3) Attempted Questions: 43 Total Questions: 50

Write the standard form of following linear programming problem:

$$Max\ Z = 5x + 4y$$

Subject to:

$$6x + 4y \leq 24$$
$$x + 2y \leq 6$$
$$-x + y \leq 1$$
$$y \leq 2$$
$$x \geq 0, y \geq 0$$

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Answer:

Write the standard form of following linear programming problem:

$$Max\ Z = 5x + 4y$$

Subject to:

$$6x + 4y \leq 24$$
$$x + 2y \leq 6$$
$$-x + y \leq 1$$
$$y \leq 2$$
$$x \geq 0, y \geq 0$$

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Answer:

$Minimize\ Z = 4x_1 + x_2$

$Subject\ to:$

$3x_1 + x_2 = 3$

$4x_1 + 3x_2 \geq 6$

$x_1 + 2x_2 \leq 4$

$x_1, x_2 \geq 0$

Write the standard form and starting tableau of the above linear programming problem.

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Answer:

$Minimize\ Z = 4x_1 + x_2$
$Subject\ to:$
$3x_1 + x_2 = 3$
$4x_1 + 3x_2 \geq 6$
$x_1 + 2x_2 \leq 4$

$x_1,\ x_2 \geq 0$

Write the standard form and starting tableau of the above linear programming problem.

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Answer:

Question: 48 (Marks: 5) Attempted Questions: 46 Total Questions: 50

Discuss why the Duality Principle is of advantageous in case of solving the following Linear Programming Problem;

Min $z = 3x_1+3x_2$

Subject to;

$2x_1+4x_2 \geq 40$

$3x_1+2x_2 \geq 50$

$x_1, x_2 \geq 0$

(Note: Solution is not required)

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Answer:

Question: 48 (Marks: 5) Attempted Questions: 46 Total Questions: 50

Discuss why the Duality Principle is of advantageous in case of solving the following Linear Programming Problem;

Min $z = 3x_1+3x_2$

Subject to;

$2x_1+4x_2 \geq 40$

$3x_1+2x_2 \geq 50$

$x_1, x_2 \geq 0$

(Note: Solution is not required)

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Answer:

Question: 49 (Marks: 5) Attempted Questions: 47 Total Questions: 50

Check whether the given initial basic feasible solution is optimal or not.

1	2	6 (7)	7
0 (10)	4	2 (2)	12
3	1 (10)	5 (1)	11
10	10	10	

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Answer:

Check whether the given initial basic feasible solution is optimal or not.

1	2	6 (7)	7
0 (10)	4	2 (2)	12
3	1 (10)	5 (1)	11
10	10	10	

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Answer:

Question: 50 (Marks: 5) Attempted Questions: 49 Total Questions: 50

A company cost Rs. 500 operations and maintenance costs are zero for the first year and increased by Rs. 100 every year. If money is worth 5% every year, calculate present worth (P(r)) for each year. The resale value of the machine is negligibly small.

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Answer:

Question: 50 (Marks: 5) Attempted Questions: 49 Total Questions: 50

A company cost Rs. 500 operations and maintenance costs are zero for the first year and increased by Rs. 100 every year. If money is worth 5% every year, calculate present worth (P(r)) for each year. The resale value of the machine is negligibly small.

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Answer:

Question: 41 (Marks: 3) Attempted Questions: 50 Total Questions: 50

Markets / salesmen	I	II	III	IV
A	44	80	52	60
B	60	56	40	72
C	36	60	48	48
D	52	76	36	40

Operate first and second operation (row & column) by optimize the above assignment problem.

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Answer:

Question: 42 (Marks: 3) Attempted Questions: 50 Total Questions: 50

The cost of a new machine is Rs. 5000. The maintenance cost during the nth year is given by M_n = Rs. 500 (n – 1), where n = 1, 2, 3, ---. If the discount rate per year is 0.05, determine discount cost ($R_t v^{t-1}$) for each year.

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.....

Answer:

Question: 43 (Marks: 3) Attempted Questions: 50 Total Questions: 50

Describe the advantages of the Duality Theory to solve a Linear Programming Problem.

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Answer:

Apply Least Cost Method to the following and determine the amount to be supplied in 1st step.

Origins	P	Q	R	Supply
A	2	7	4	3
B	3	3	1	4
C	5	4	1	5
Demand	4	2	6	

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Answer:

Question: 45 (Marks: 3) Attempted Questions: 50 Total Questions: 50

Write the standard form of following linear programming problem:

$$Max\ Z = 5x + 4y$$

Subject to:

$$6x + 4y \leq 24$$
$$x + 2y \leq 6$$
$$-x + y \leq 1$$
$$y \leq 2$$
$$x \geq 0, y \geq 0$$

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Answer:

Question: 46 (Marks: 3) Attempted Questions: 50 Total Questions: 50

$Minimize\ Z = 4x_1 + x_2$
$Subject\ to:$
$3x_1 + x_2 = 3$
$4x_1 + 3x_2 \geq 6$
$x_1 + 2x_2 \leq 4$

$x_1,\ x_2 \geq 0$

Write the standard form and starting tableau of the above linear programming problem.

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Answer:

Question: 47 (Marks: 5) Attempted Questions: 50 Total Questions: 50

By using two phase method, express the first phase of the following linear programming problem in terms non-basic variables.

$$\text{Minimize } Z = 3x_1 + 2x_2 + 4x_3$$

Subject to

$$2x_1 + x_2 + 3x_3 = 60$$

$$3x_1 + 3x_2 + 5x_3 \geq 120$$

$$x_1, x_2, x_3 \geq 0$$

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Answer:

Discuss why the Duality Principle is of advantageous in case of solving the following
Linear Programming Problem;
Min $z = 3x_1+3x_2$
Subject to;
$2x_1+4x_2 \geq 40$
$3x_1+2x_2 \geq 50$
$x_1, x_2 \geq 0$
(**Note**: Solution is not required)

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Answer:

]
a

Question: 49 (Marks: 5) Attempted Questions: 50 Total Questions: 50

Check whether the given initial basic feasible solution is optimal or not.

1	2	6 (7)	7
0 (10)	4	2 (2)	12
3	1 (10)	5 (1)	11
10	10	10	

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Answer:

Question: 50 (Marks: 5) Attempted Questions: 50 Total Questions: 50

A company cost Rs. 500 operations and maintenance costs are zero for the first year and increased by Rs. 100 every year. If money is worth 5% every year, calculate present worth (P(r)) for each year. The resale value of the machine is negligibly small.

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Answer: